

# FASTA format

In bioinformatics and biochemistry, the **FASTA format** is a text-based format for representing either nucleotide sequences or amino acid (protein) sequences, in which nucleotides or amino acids are represented using single-letter codes.

The format allows for sequence names and comments to precede the sequences. It originated from the FASTA software package and has since become a near-universal standard in bioinformatics.<sup>[4]</sup>

The simplicity of FASTA format makes it easy to manipulate and parse sequences using text-processing tools and scripting languages.

## Overview

A sequence begins with a greater-than character (">") followed by a description of the sequence (all in a single line). The lines immediately following the description line are the sequence representation, with one letter per amino acid or nucleic acid, and are typically no more than 80 characters in length.

For example:

```
>MCHU - Calmodulin - Human, rabbit, bovine, rat, and
chicken
MADQLTEEQIAEFKEAFSLFDKDGDTITTKELGTVMRSLGQNPTAEALQDMINEVD
ADGNGTID
FPEFLTMMARKMKDSTDSEEEIREAFRVFDKDGNGYISAAELRHVMTNLGEKLTDEEV
DEMIREA
DIDGDGQVNYEEFVQMMTAK*
```

## Original format

The original FASTA/Pearson format is described in the documentation for the FASTA suite of programs. It can be downloaded with any free distribution of FASTA (see fasta20.doc, fastaVN.doc, or fastaVN.me—where VN is the Version Number).

In the original format, a sequence was represented as a series of lines, each of which was no longer than 120 characters and usually did not exceed 80 characters. This probably was to allow for the preallocation of fixed line sizes in software: at the time most users relied on Digital Equipment Corporation (DEC) VT220 (or compatible) terminals which could display 80 or 132 characters per line. Most people preferred the bigger font in 80-character modes and so it became the recommended fashion to use 80 characters or less (often 70) in FASTA lines. Also, the width of a standard printed page is 70 to 80 characters (depending on the font). Hence, 80 characters became the norm.

### FASTA format

|                                      |                                                                                                         |
|--------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Filename extensions</b>           | .fasta, .fas, .fa, .fna, .ffn, .faa, .mpfa, .frn                                                        |
| <b>Internet media type</b>           | text/x-fasta                                                                                            |
| <b>Uniform Type Identifier (UTI)</b> | no                                                                                                      |
| <b>Developed by</b>                  | <u>David J. Lipman</u><br><u>William R. Pearson</u> <sup>[1][2]</sup>                                   |
| <b>Initial release</b>               | 1985                                                                                                    |
| <b>Type of format</b>                | <u>Bioinformatics</u>                                                                                   |
| <b>Extended from</b>                 | <u>ASCII for FASTA</u>                                                                                  |
| <b>Extended to</b>                   | <u>FASTQ format</u> <sup>[3]</sup>                                                                      |
| <b>Website</b>                       | <u>www.ncbi.nlm.nih.gov/BLAST/fasta.shtml</u> ( <u>https://www.ncbi.nlm.nih.gov/BLAST/fasta.shtml</u> ) |

The first line in a FASTA file started either with a ">" (greater-than) symbol or, less frequently, a ";" (semicolon) was taken as a comment. Subsequent lines starting with a semicolon would be ignored by software. Since the only comment used was the first, it quickly became used to hold a summary description of the sequence, often starting with a unique library accession number, and with time it has become commonplace to always use ">" for the first line and to not use ";" comments (which would otherwise be ignored).

Following the initial line (used for a unique description of the sequence) was the actual sequence itself in the standard one-letter character string. Anything other than a valid character would be ignored (including spaces, tabulators, asterisks, etc...). It was also common to end the sequence with an "\*" (asterisk) character (in analogy with use in PIR formatted sequences) and, for the same reason, to leave a blank line between the description and the sequence. Below are a few sample sequences:

```
;LCB0 - Prolactin precursor - Bovine
; a sample sequence in FASTA format
MDSKGSSQKGSRLLLLLLVSNLLLCQGVVSTPVCNPGNCQVSLRDLFDRAVMVSHYIHDLS
EMFNEFDKRYAQKGFFITMALNSCHTSSLPTPEDKEQAQQTHHEVLMSLILGLLRSWNDPLYHL
VTEVRGMKGAPDAILSRATEIEEENKRLLEGMEMIFGQVIPGAKETEPYPVWSGLPSLQTKDED
ARYSAFYNLLHCLRRDSSKIDTYLKLNCRIIYNNNC*

>MCHU - Calmodulin - Human, rabbit, bovine, rat, and chicken
MADQLTEEQIAEFKEAFSLFDKDGDTITTKELGTVMRSLGQNPTEAELQDMINEVDADGNGTID
FPEFLTMMARKMKDSTDSEEEIREAFRVFDKDGNGYISAAELRHVMTNLGEKLTDEEVDEMIREA
DIDGQGQVNYEEFVQMMTAK*

>gi|5524211|gb|AAD44166.1| cytochrome b [Elephas maximus maximus]
LCLYTHIGRNIYYGSYLYSETWNTGIMLLITMATAFMGYVLPWGQMSFWGATVITNLFSAIPYIGTNLV
EWIWGGFSVDKATLNRFFAFHFILPFTMVALAGVHLTFLHETGSNNPLGLTSDSDKIPFHPYYTIKDFLG
LLILILLLLLLLALLSPDMLGDPDNHMPADPLNTPLHIKPEWYFLFAYAILRSVPNKLGGVLALFLSIVIL
GLMPFLHTSKHRSMMLRPLSQALFWTLTMDLLTLTWIGSQPVEYPYTIIGQMASILYFSIILAFPLIAGX
IENY
```

A multiple-sequence FASTA format, or multi-FASTA format, would be obtained by concatenating several single-sequence FASTA files in one file. This does not imply a contradiction with the format as only the first line in a FASTA file may start with a ";" or ">", forcing all subsequent sequences to start with a ">" in order to be taken as separate sequences (and further forcing the exclusive reservation of ">" for the sequence definition line). Thus, the examples above would be a multi-FASTA file if taken together.

Modern bioinformatics programs that rely on the FASTA format expect the sequence headers to be preceded by ">". The sequence is generally represented as "interleaved", or on multiple lines as in the above example, but may also be "sequential", or on a single line. Running different bioinformatics programs may require conversions between "sequential" and "interleaved" FASTA formats.

## Description line

The description line (define) or header/identifier line, which begins with ">", gives a name and/or a unique identifier for the sequence, and may also contain additional information. In a deprecated practice, the header line sometimes contained more than one header, separated by a ^A (Control-A) character. In the original Pearson FASTA format, one or more comments, distinguished by a semi-colon at the beginning of the line, may occur after the header. Some databases and bioinformatics applications do not recognize these comments and follow the NCBI FASTA specification (<https://www.ncbi.nlm.nih.gov/blast/fasta.shtml>). An example of a multiple sequence FASTA file follows:

```
>SEQUENCE_1
MTEITAAMVKELRESTGAGMMDCKNALSETNGDFDKAVQLLREKGLGKAACKADRLAAEG
LVSVKVSDDFTIAAMRPSYLSYEDLDMTFVENEYKALVAELEKENEERRRLKDPNKPEHK
IPQFASRKQLSDAILKEAEKIKEELKAQGKPEKIWDNIIPGKMNSFIADNSQLDSKLTLL
```

```
MGQFYVMDDKKTVEQVIAEKEKEFGGKIKIVEFICFEVGEGLEKKTEDFAAEVAAQL  
>SEQUENCE_2  
SATVSEINSETDFVAKNDQFIALTKDITTAHIQSNSLQSVEELHSSTINGVKFEEYLKSQI  
ATIGENLVVRRFATLKAGANGVVNGYIHTNGRVGVVIAAACDSAEVASKSRDLLRQICMH
```

## NCBI identifiers

The NCBI defined a standard for the unique identifier used for the sequence (SeqID) in the header line. This allows a sequence that was obtained from a database to be labelled with a reference to its database record. The database identifier format is understood by the NCBI tools like `makeblastdb` and `table2asn`. The following list describes the NCBI FASTA defined format for sequence identifiers.<sup>[5]</sup>

| Type                                                                                                                                                              | Format(s)                                                                     | Example(s)                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------|
| local (i.e. no database reference)                                                                                                                                | lcl  <i>integer</i><br>lcl  <i>string</i>                                     | lcl 123<br>lcl hmm271           |
| GenInfo backbone seqid                                                                                                                                            | bbs  <i>integer</i>                                                           | bbs 123                         |
| GenInfo backbone moltype                                                                                                                                          | bbm  <i>integer</i>                                                           | bbm 123                         |
| GenInfo import ID                                                                                                                                                 | gim  <i>integer</i>                                                           | gim 123                         |
| GenBank ( <a href="https://www.ncbi.nlm.nih.gov/Genbank/index.html">https://www.ncbi.nlm.nih.gov/Genbank/index.html</a> )                                         | gb  <i>accession</i>   <i>locus</i>                                           | gb M73307 AGMA13GT              |
| EMBL ( <a href="http://www.embl-heidelberg.de">http://www.embl-heidelberg.de</a> )                                                                                | emb  <i>accession</i>   <i>locus</i>                                          | emb CAM43271.1                  |
| PIR ( <a href="https://web.archive.org/web/20140312021627/http://pir.georgetown.edu/">https://web.archive.org/web/20140312021627/http://pir.georgetown.edu/</a> ) | pir  <i>accession</i>   <i>name</i>                                           | pir  G36364                     |
| SWISS-PROT ( <a href="http://www.ebi.ac.uk/swissprot">http://www.ebi.ac.uk/swissprot</a> )                                                                        | sp  <i>accession</i>   <i>name</i>                                            | sp P01013 OVAX_CHICK            |
| patent                                                                                                                                                            | pat  <i>country</i>   <i>patent</i>   <i>sequence-number</i>                  | pat US RE33188 1                |
| pre-grant patent                                                                                                                                                  | pgp  <i>country</i>   <i>application-number</i>   <i>sequence-number</i>      | pgp EP 0238993 7                |
| RefSeq ( <a href="https://www.ncbi.nlm.nih.gov/projects/RefSeq">https://www.ncbi.nlm.nih.gov/projects/RefSeq</a> )                                                | ref  <i>accession</i>   <i>name</i>                                           | ref NM_010450.1                 |
| general database reference<br>(a reference to a database that's not in this list)                                                                                 | gnl  <i>database</i>   <i>integer</i><br>gnl  <i>database</i>   <i>string</i> | gnl taxon 9606<br>gnl PID e1632 |
| GenInfo integrated database                                                                                                                                       | gi  <i>integer</i>                                                            | gi 21434723                     |
| DDBJ ( <a href="http://www.ddbj.nig.ac.jp">http://www.ddbj.nig.ac.jp</a> )                                                                                        | dbj  <i>accession</i>   <i>locus</i>                                          | dbj BAC85684.1                  |
| PRF ( <a href="http://www.prf.or.jp">http://www.prf.or.jp</a> )                                                                                                   | prf  <i>accession</i>   <i>name</i>                                           | prf  0806162C                   |
| PDB ( <a href="https://web.archive.org/web/20080828002005/http://www.rcsb.org/pdb">https://web.archive.org/web/20080828002005/http://www.rcsb.org/pdb</a> )       | pdb  <i>entry</i>   <i>chain</i>                                              | pdb 1I4L D                      |
| third-party GenBank ( <a href="https://www.ncbi.nlm.nih.gov/Genbank/index.html">https://www.ncbi.nlm.nih.gov/Genbank/index.html</a> )                             | tpg  <i>accession</i>   <i>name</i>                                           | tpg BK003456                    |
| third-party EMBL ( <a href="http://www.embl-heidelberg.de">http://www.embl-heidelberg.de</a> )                                                                    | tpe  <i>accession</i>   <i>name</i>                                           | tpe BN000123                    |
| third-party DDBJ ( <a href="http://www.ddbj.nig.ac.jp">http://www.ddbj.nig.ac.jp</a> )                                                                            | tpd  <i>accession</i>   <i>name</i>                                           | tpd FAA00017                    |
| TrEMBL                                                                                                                                                            | tr  <i>accession</i>   <i>name</i>                                            | tr Q90RT2 Q90RT2_9HIV1          |

The vertical bars ("|") in the above list are not separators in the sense of the Backus–Naur form but are part of the format. Multiple identifiers can be concatenated, also separated by vertical bars.

## Sequence representation

---

Following the header line, the actual sequence is represented. Sequences may be protein sequences or nucleic acid sequences, and they can contain gaps or alignment characters (see sequence alignment). Sequences are expected to be represented in the standard IUB/IUPAC amino acid and nucleic acid codes, with these exceptions: lower-case letters are accepted and are mapped into upper-case; a single hyphen or dash can be used to represent a gap character; and in amino acid sequences, U and \* are acceptable letters (see below). Numerical digits are not allowed but are used in some databases to indicate the position in the sequence. The nucleic acid codes supported are:<sup>[6][7][8]</sup>

| Nucleic Acid Code | Meaning                          | Mnemonic                        |
|-------------------|----------------------------------|---------------------------------|
| A                 | A                                | <u>A</u> denine                 |
| C                 | C                                | <u>C</u> ytosine                |
| G                 | G                                | <u>G</u> uanine                 |
| T                 | T                                | <u>T</u> hymine                 |
| U                 | U                                | <u>U</u> racil                  |
| (i)               | i                                | <u>i</u> nosine (non-standard)  |
| R                 | A or G (I)                       | <u>puR</u> ine                  |
| Y                 | C, T or U                        | <u>pY</u> rimidines             |
| K                 | G, T or U                        | bases which are <u>K</u> etones |
| M                 | A or C                           | bases with <u>aM</u> ino groups |
| S                 | C or G                           | <b>S</b> trong interaction      |
| W                 | A, T or U                        | <b>W</b> eak interaction        |
| B                 | not A (i.e. C, G, T or U)        | <b>B</b> comes after A          |
| D                 | not C (i.e. A, G, T or U)        | <b>D</b> comes after C          |
| H                 | not G (i.e., A, C, T or U)       | <b>H</b> comes after G          |
| V                 | neither T nor U (i.e. A, C or G) | <b>V</b> comes after U          |
| N                 | A C G T U                        | <b>N</b> ucleic acid            |
| -                 | gap of indeterminate length      |                                 |

The amino acid codes supported (22 amino acids and 3 special codes) are:

| Amino Acid Code | Meaning                                           |
|-----------------|---------------------------------------------------|
| A               | <u>Alanine</u>                                    |
| B               | <u>Aspartic acid</u> (D) or <u>Asparagine</u> (N) |
| C               | <u>Cysteine</u>                                   |
| D               | <u>Aspartic acid</u>                              |
| E               | <u>Glutamic acid</u>                              |

|   |                                           |
|---|-------------------------------------------|
| F | <u>Phenylalanine</u>                      |
| G | <u>Glycine</u>                            |
| H | <u>Histidine</u>                          |
| I | <u>Isoleucine</u>                         |
| J | <u>Leucine (L) or Isoleucine (I)</u>      |
| K | <u>Lysine</u>                             |
| L | <u>Leucine</u>                            |
| M | <u>Methionine/Start codon</u>             |
| N | <u>Asparagine</u>                         |
| O | <u>Pyrrolysine (rare)</u>                 |
| P | <u>Proline</u>                            |
| Q | <u>Glutamine</u>                          |
| R | <u>Arginine</u>                           |
| S | <u>Serine</u>                             |
| T | <u>Threonine</u>                          |
| U | <u>Selenocysteine (rare)</u>              |
| V | <u>Valine</u>                             |
| W | <u>Tryptophan</u>                         |
| Y | <u>Tyrosine</u>                           |
| Z | <u>Glutamic acid (E) or Glutamine (Q)</u> |
| X | any                                       |
| * | translation stop                          |
| - | gap of indeterminate length               |

## FASTA file

---

### Filename extension

There is no standard filename extension for a text file containing FASTA formatted sequences. The table below shows each extension and its respective meaning.

| Extension                     | Meaning                          | Notes                                                         |
|-------------------------------|----------------------------------|---------------------------------------------------------------|
| fasta, fas, fa <sup>[9]</sup> | generic FASTA                    | Any generic FASTA file                                        |
| fna                           | FASTA nucleic acid               | Used generically to specify nucleic acids                     |
| ffn                           | FASTA nucleotide of gene regions | Contains coding regions for a genome                          |
| faa                           | FASTA amino acid                 | Contains amino acid sequences                                 |
| mpfa                          | FASTA amino acids                | Contains multiple protein sequences                           |
| frn                           | FASTA <u>non-coding RNA</u>      | Contains non-coding RNA regions for a genome, e.g. tRNA, rRNA |

## Compression

The compression of FASTA files requires a specific compressor to handle both channels of information: identifiers and sequence. For improved compression results, these are mainly divided into two streams where the compression is made assuming independence. For example, the algorithm MFCompress<sup>[10]</sup> performs lossless compression of these files using context modelling and arithmetic encoding. Genozip,<sup>[11]</sup> a software package for compressing genomic files, uses an extensible context-based model. Benchmarks of FASTA file compression algorithms have been reported by Hosseini et al. in 2016,<sup>[12]</sup> and Kryukov et al. in 2020.<sup>[13]</sup>

## Encryption

The encryption of FASTA files can be performed with various tools, including Cryfa and Genozip. Cryfa uses AES encryption and also enables data compression.<sup>[14][15]</sup> Similarly, Genozip can encrypt FASTA files with AES-256 during compression.<sup>[11]</sup>

## Extensions

---

FASTQ format is a form of FASTA format extended to indicate information related to sequencing. It is created by the Sanger Centre in Cambridge.<sup>[3]</sup>

A2M/A3M are a family of FASTA-derived formats used for sequence alignments. In A2M/A3M sequences, lowercase characters are taken to mean insertions, which are then indicated in the other sequences as the dot (".") character. The dots can be discarded for compactness without loss of information. As with typical FASTA files used in alignments, the gap ("-") is taken to mean exactly one position.<sup>[16]</sup> A3M is similar to A2M, with the added rule that gaps aligned to insertions can too be discarded.<sup>[17]</sup>

## Working with FASTA files

---

A plethora of user-friendly scripts are available from the community to perform FASTA file manipulations. Online toolboxes, such as FaBox<sup>[18]</sup> or the FASTX-Toolkit within Galaxy servers, are also available.<sup>[19]</sup> These can be used to segregate sequence headers/identifiers, rename them, shorten them, or extract sequences of interest from large FASTA files based on a list of wanted identifiers (among other available



functions). A tree-based approach to sorting multi-FASTA files (TREE2FASTA<sup>[20]</sup>) also exists based on the coloring and/or annotation of sequences of interest in the FigTree viewer. Additionally, the Bioconductor *Biostrings* package can be used to read and manipulate FASTA files in R.<sup>[21]</sup>

Several online format converters exist to rapidly reformat multi-FASTA files to different formats (e.g. NEXUS, PHYLIP) for use with different phylogenetic programs, such as the converter available on phylogeny.fr.<sup>[22]</sup>

## See also

---

- The FASTQ format, used to represent DNA sequencer reads along with quality scores.
- The SAM and CRAM formats, used to represent genome sequencer reads that have been aligned to genome sequences.
- The GVF format (Genome Variation Format), an extension based on the GFF3 format.

## References

---

1. Lipman DJ, Pearson WR (March 1985). "Rapid and sensitive protein similarity searches". *Science*. **227** (4693): 1435–41. Bibcode:1985Sci...227.1435L (<https://ui.adsabs.harvard.edu/abs/1985Sci...227.1435L>). doi:10.1126/science.2983426 (<https://doi.org/10.1126%2Fscience.2983426>). PMID 2983426 (<https://pubmed.ncbi.nlm.nih.gov/2983426>). 
2. Pearson WR, Lipman DJ (April 1988). "Improved tools for biological sequence comparison" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC280013>). *Proceedings of the National Academy of Sciences of the United States of America*. **85** (8): 2444–8. Bibcode:1988PNAS...85.2444P (<https://ui.adsabs.harvard.edu/abs/1988PNAS...85.2444P>). doi:10.1073/pnas.85.8.2444 (<https://doi.org/10.1073%2Fpnas.85.8.2444>). PMC 280013 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC280013>). PMID 3162770 (<https://pubmed.ncbi.nlm.nih.gov/3162770>).
3. Cock PJ, Fields CJ, Goto N, Heuer ML, Rice PM (April 2010). "The Sanger FASTQ file format for sequences with quality scores, and the Solexa/Illumina FASTQ variants" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2847217>). *Nucleic Acids Research*. **38** (6): 1767–71. doi:10.1093/nar/gkp1137 (<https://doi.org/10.1093%2Fnar%2Fgkp1137>). PMC 2847217 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2847217>). PMID 20015970 (<https://pubmed.ncbi.nlm.nih.gov/20015970>).
4. "What is FASTA format?" (<https://zhanggroup.org/FASTA/>). *Zhang Lab*. Archived (<https://web.archive.org/web/20221204183844/https://zhanggroup.org/FASTA/>) from the original on 2022-12-04. Retrieved 2022-12-04.
5. *NCBI C++ Toolkit Book* ([https://ncbi.github.io/cxx-toolkit/pages/ch\\_demo#ch\\_demo.id1\\_fetch.html\\_ref\\_fasta](https://ncbi.github.io/cxx-toolkit/pages/ch_demo#ch_demo.id1_fetch.html_ref_fasta)). National Center for Biotechnology Information. Retrieved 2018-12-19.
6. Tao Tao (2011-08-24). "Single Letter Codes for Nucleotides" ([https://web.archive.org/web/20120914234405/http://www.ncbi.nlm.nih.gov/staff/tao/tools/tool\\_lettercode.html](https://web.archive.org/web/20120914234405/http://www.ncbi.nlm.nih.gov/staff/tao/tools/tool_lettercode.html)). [*NCBI Learning Center*]. National Center for Biotechnology Information. Archived from the original ([https://www.ncbi.nlm.nih.gov/staff/tao/tools/tool\\_lettercode.html](https://www.ncbi.nlm.nih.gov/staff/tao/tools/tool_lettercode.html)) on 2012-09-14. Retrieved 2012-03-15.
7. "IUPAC code table" (<https://web.archive.org/web/20110811073845/http://www.dna.affrc.go.jp/misc/MPsrch/InfoIUPAC.html>). NIAS DNA Bank. Archived from the original (<http://www.dna.affrc.go.jp/misc/MPsrch/InfoIUPAC.html>) on 2011-08-11.
8. "anysymbol" (<https://mafft.cbrc.jp/alignment/software/anysymbol.html>). *MAFFT - a multiple sequence alignment program*.

9. "Alignment Fileformats" (<http://www.jalview.org/help/html/io/fileformats.html>). 22 May 2019. Retrieved 22 May 2019.
10. Pinho AJ, Pratas D (January 2014). "MFCompress: a compression tool for FASTA and multi-FASTA data" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3866555>). *Bioinformatics*. **30** (1): 117–8. doi:10.1093/bioinformatics/btt594 (<https://doi.org/10.1093%2Fbioinformatics%2Fbtt594>). PMC 3866555 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3866555>). PMID 24132931 (<https://pubmed.ncbi.nlm.nih.gov/24132931>).
11. Lan, Divon; Tobler, Ray; Souilmi, Yassine; Llamas, Bastien (2021-02-15). "Genozip: a universal extensible genomic data compressor" (<https://doi.org/10.1093/bioinformatics/btab102>). *Bioinformatics*. **37** (16): 2225–2230. doi:10.1093/bioinformatics/btab102 (<https://doi.org/10.1093%2Fbioinformatics%2Fbtab102>). ISSN 1367-4803 (<https://www.worldcat.org/issn/1367-4803>). PMC 8388020 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8388020>). PMID 33585897 (<https://pubmed.ncbi.nlm.nih.gov/33585897>).
12. Hosseini, Morteza; Pratas, Diogo; Pinho, Armando J. (2016). "A Survey on Data Compression Methods for Biological Sequences" (<https://doi.org/10.3390/info7040056>). *Information*. **7** (4): 56. doi:10.3390/info7040056 (<https://doi.org/10.3390%2Finfo7040056>). ISSN 2078-2489 (<https://www.worldcat.org/issn/2078-2489>).
13. Kryukov K, Ueda MT, Nakagawa S, Imanishi T (July 2020). "Sequence Compression Benchmark (SCB) database—A comprehensive evaluation of reference-free compressors for FASTA-formatted sequences" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7336184>). *GigaScience*. **9** (7): gaaa072. doi:10.1093/gigascience/gaaa072 (<https://doi.org/10.1093%2Fgigascience%2Fgaaa072>). PMC 7336184 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7336184>). PMID 32627830 (<https://pubmed.ncbi.nlm.nih.gov/32627830>).
14. Pratas D, Hosseini M, Pinho A (2017). "Cryfa: a tool to compact and encrypt FASTA files". *11th International Conference on Practical Applications of Computational Biology & Bioinformatics (PACBB)*. Advances in Intelligent Systems and Computing. Vol. 616. Springer. pp. 305–312. doi:10.1007/978-3-319-60816-7\_37 ([https://doi.org/10.1007%2F978-3-319-60816-7\\_37](https://doi.org/10.1007%2F978-3-319-60816-7_37)). ISBN 978-3-319-60815-0.
15. Hosseini, Morteza; Pratas, Diogo; Pinho, Armando J (2019-01-01). Berger, Bonnie (ed.). "Cryfa: a secure encryption tool for genomic data" (<https://academic.oup.com/bioinformatics/article/35/1/146/5055587>). *Bioinformatics*. **35** (1): 146–148. doi:10.1093/bioinformatics/bty645 (<https://doi.org/10.1093%2Fbioinformatics%2Fbty645>). ISSN 1367-4803 (<https://www.worldcat.org/issn/1367-4803>). PMC 6298042 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6298042>). PMID 30020420 (<https://pubmed.ncbi.nlm.nih.gov/30020420>).
16. "Description of A2M alignment format" (<https://web.archive.org/web/20220815104407/https://compbio.soe.ucsc.edu/a2m-desc.html>). *SAMtools*. Archived from the original (<https://compbio.soe.ucsc.edu/a2m-desc.html>) on 2022-08-15.
17. "soedinglab/hh-suite: reformat.pl" (<https://github.com/soedinglab/hh-suite/blob/master/scripts/reformat.pl>). *GitHub*. 20 November 2022.
18. Villesen, P. (2007). "FaBox: an online toolbox for fasta sequences" (<https://onlinelibrary.wiley.com/doi/10.1111/j.1471-8286.2007.01821.x>). *Molecular Ecology Notes*. **7** (6): 965–968. doi:10.1111/j.1471-8286.2007.01821.x (<https://doi.org/10.1111%2Fj.1471-8286.2007.01821.x>). ISSN 1471-8278 (<https://www.worldcat.org/issn/1471-8278>).
19. Blankenberg D, Von Kuster G, Bouvier E, Baker D, Afgan E, Stoler N, Galaxy Team, Taylor J, Nekrutenko A (2014). "Dissemination of scientific software with Galaxy ToolShed" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4038738>). *Genome Biology*. **15** (2): 403. doi:10.1186/gb4161 (<https://doi.org/10.1186%2Fgb4161>). PMC 4038738 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4038738>). PMID 25001293 (<https://pubmed.ncbi.nlm.nih.gov/25001293>).

20. Sauvage T, Plouviez S, Schmidt WE, Fredericq S (March 2018). "TREE2FASTA: a flexible Perl script for batch extraction of FASTA sequences from exploratory phylogenetic trees" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5838971>). *BMC Research Notes*. **11** (1): 403. doi:10.1186/s13104-018-3268-y (<https://doi.org/10.1186%2Fs13104-018-3268-y>). PMC 5838971 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5838971>). PMID 29506565 (<https://pubmed.ncbi.nlm.nih.gov/29506565>).
21. Pagès, H; Aboyoun, P; Gentleman, R; DebRoy, S (2018). "*Biostrings: Efficient manipulation of biological strings*" (<https://bioconductor.org/packages/release/bioc/html/Biostrings.html>). *Bioconductor.org*. R package version 2.48.0. doi:10.18129/B9.bioc.Biostrings (<https://doi.org/10.18129%2FB9.bioc.Biostrings>).
22. Dereeper A, Guignon V, Blanc G, Audic S, Buffet S, Chevenet F, Dufayard JF, Guindon S, Lefort V, Lescot M, Claverie JM, Gascuel O (July 2008). "Phylogeny.fr: robust phylogenetic analysis for the non-specialist" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2447785>). *Nucleic Acids Research*. **36** (Web Server issue): W465–9. doi:10.1093/nar/gkn180 (<https://doi.org/10.1093%2Fnar%2Fgkn180>). PMC 2447785 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2447785>). PMID 18424797 (<https://pubmed.ncbi.nlm.nih.gov/18424797>).

## External links

---

- Bioconductor (<http://www.bioconductor.org>)
  - FASTX-Toolkit ([http://hannonlab.cshl.edu/fastx\\_toolkit/](http://hannonlab.cshl.edu/fastx_toolkit/))
  - FigTree viewer (<http://tree.bio.ed.ac.uk/software/figtree/>)
  - Phylogeny.fr ([http://phylogeny.lirmm.fr/phylo\\_cgi/data\\_converter.cgi](http://phylogeny.lirmm.fr/phylo_cgi/data_converter.cgi))
  - GTO (<http://bioinformatics.ua.pt/gto>)
- 

Retrieved from "[https://en.wikipedia.org/w/index.php?title=FASTA\\_format&oldid=1184474058](https://en.wikipedia.org/w/index.php?title=FASTA_format&oldid=1184474058)"

▪